Project Two: Modeling Juvenile Delinquency

October 2026

1 Introduction

Gangs are defined by the Department of Justice to be "associations of three or more individuals who adopt a group identity in order to create an atmosphere of fear or intimidation," [4] and are among the top drivers of violent crime in urban areas. In Memphis, TN especially, gang violence has historically contributed to nationally outstanding violent and drug-related crime rates. The involvement of juveniles, aged 13 to 17, has been a particular issue within gangs. It is estimated that around 40 % of gang members are under the age of 18, putting these juveniles at risk of perpetrating and falling victim to violent crime [6].

A model proposed by Lee and Do [2], displayed the spread of gang membership and delinquency due to peer pressure among at-risk juveniles to be similar to the spread of a disease. We seek to produce a similar model of gang-related juvenile delinquency based on data obtained by Lee and Do [2] and on data obtained for Memphis, modeling involvement with gang activity as similar to the spread of a contagious disease. In constructing this model, we hope to better understand the dynamics of gang activity among vulnerable juveniles.

We also consider a number of initiatives which have been taken in the last few years to reduce gang activity in Memphis, particularly among juveniles. These initiatives fall broadly into two categories: aggressive law enforcement campaigns and prevention-focused youth outreach programs. In 2025, the Memphis Safe task force was formed to address the high prevalence of gang-related violent crime in Memphis. This is an example of aggressive efforts by law enforcement to reduce crime by increasing arrests and incarcerations. The Memphis-Shelby County school district has also implemented youth outreach programs such as Gang Resistance For Saving Society's Youth (G.R.A.S.S.Y.) which involves a mixture of youth intervention, education, and employment initiatives to reduce early gang involvement. In modeling these prevention efforts, we seek to evaluate their effectiveness and their overall impact on trends of gang violence in juveniles.

2 Methods

2.1 System Dynamics Overview

We base our model on the SARS model outlined in section 4.3 of Introduction to Computational Science [7]. In keeping with the model proposed by Lee and Do (2011), we include the following populations of at-risk juveniles: susceptible teens (S) who are not associated with or actively participating in gang violence, non-delinquent gang members (G) who are affiliated with gangs but not actively involved in criminal activity, delinquent gang members (D) who are involved with criminal activity but have not been detained by law enforcement, and law-enforced members (L) who have been apprehended by law enforcement and have no contact with other gang members.

A member of S can become influenced to join a gang G through the influence of existing gang members. Furthermore, members of G may become members of D through a mixture of peer-pressure and personal issues. Members of G also occasionally return to S through good influences or fear. A certain number of delinquent gang members are apprehended by law enforcement and incarcerated before being released to S, G, or D.

For the sake of our model, we assume that all juveniles in Memphis belongs to exactly one population at a time and are susceptible to gang activity, all are equally susceptible regardless of gender or socioeconomic status, and juveniles are only influenced to participate in gang activity by other juveniles. Juveniles in the law-enforced population are assumed to have no contact with the other groups while detained. Additionally, gang involvement is assumed to be influenced only by interactions between juveniles. The total juvenile population (N) is assumed to remain constant throughout the simulation. All incoming juveniles enter the susceptible population, while individuals may age out of any of the four groups at a constant rate. The overall rates of aging in and out are equal, maintaining a constant population despite changes in the individual groups.

2.2 Mathematical Model

We model the transitions between the four juvenile populations using the following system of differential equations:

$$\frac{dS}{dt} = mN + r_G G + r_L L - \frac{j_G S(G+D)}{N} - mS \tag{1}$$

$$\frac{dG}{dt} = \frac{j_G S(G+D)}{N} + r_L L - g_p G - \frac{g_i GD}{N} - r_G G - mG \tag{2}$$

$$\frac{dD}{dt} = g_p G + \frac{g_i GD}{N} + r_L L - iD - mD \tag{3}$$

$$\frac{dL}{dt} = iD - 3r_L L - mL \tag{4}$$

Where the total population is

$$N = S + G + D + L.$$

The parameters are

$$\begin{aligned} j_G &= 0.60, & g_p &= 0.14, \\ g_i &= 1.00, & r_G &= 0.30, \\ r_L &= 1.00, & i &= 0.235, \\ m &= 0.20. \end{aligned}$$

Where r_G is the rate at which non-delinquent gang members return to being susceptible, r_L is the rate at which incarcerated juveniles return to the susceptible group, j_G is the rate at which susceptible juveniles are influenced to join gangs, g_p is the rate at which gang members become delinquent due to personal reasons, gi is the rate at which gang members become delinquent due to influence, and i is the rate at which delinquent gang members become incarcerated.

We may also account for efforts to curb juvenile gang activity in our model, in particular, we examine efforts from Memphis-Shelby County Schools and the Memphis Safe Task Force to reduce gang activity. The G.R.A.S.S.Y. program, an early prevention program aimed at protecting adolescents from gang influence, is active in many Memphis schools. We model this prevention as a kind of vaccination against gang activity, creating a new population of students who have been reached by intervention methods, denoted V. We assume in modeling this prevention program that students influenced by it are entirely immune to factors that might otherwise influence them to gang activity. A recent effort to curb gang activity is the Memphis Safe Task Force, which has cracked down hard against violent crime in 2025 and 2026, claiming to have arrested 1,461 gang members as of September 2026 [5]. To model the increase, we increase the incarceration rate for delinquent gang members to model the increase in arrests reported in the first year of the task force's operation. These changes produce the following system of equations.

$$\frac{dS}{dt} = mN + r_G G + r_L L - \frac{j_G S(G+D)}{N} - vS - mS \tag{5}$$

$$\frac{dV}{dt} = vS - mV \tag{6}$$

$$\frac{dG}{dt} = \frac{j_G S(G+D)}{N} + r_L L - g_p G - \frac{g_i GD}{N} - r_G G - mG \tag{7}$$

$$\frac{dD}{dt} = g_p G + \frac{g_i GD}{N} + r_L L - i_1 D - mD \tag{8}$$

$$\frac{dL}{dt} = i_1 D - 3r_L L - mL \tag{9}$$

In this extended model, the total population is

$$N = S + V + G + D + L.$$

The parameters are the same as the initial model except for these additions:

$$v = 0.1$$
$$i_1 = 0.296$$

Where v is the number of juveniles influenced away from gang activity who are no longer susceptible, and i_1 is the increased incarceration rate. Data from Memphis Police Department from 2024 reports a total of 19,000 arrests [3] while data from 2026 reports that the Memphis Safe Task Force has made a total of 23,000 arrests during its first year of operation [5]. This represents a 21% increase in arrests, which has been applied to the initial incarceration rate accordingly.

3 Results

3.1 Initial Model

We first simulated the juvenile delinquency model using the initial population proportions and transition rates described in Section 2. The simulation was conducted for 100 years using the Forward Euler method to examine the long-term behavior of the susceptible (S), non-delinquent gang (G), delinquent gang (D), and law-enforced (L) populations.

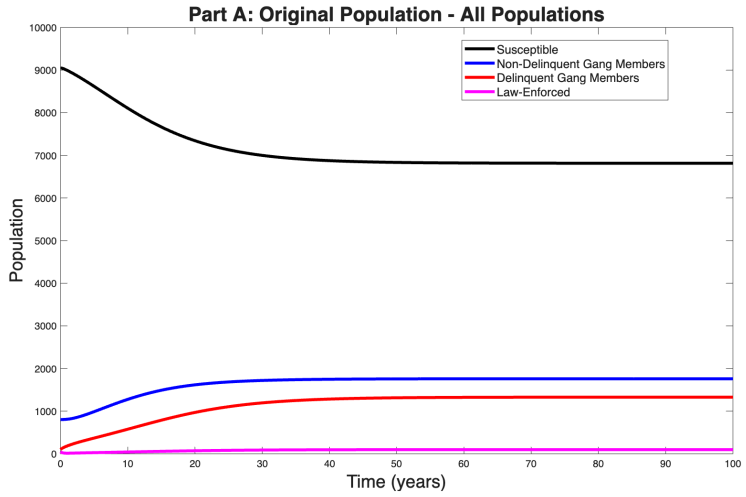


Figure 1: Population of susceptible and gang-affiliated juveniles in the initial model using the original parameters from Lee and Do (2011)

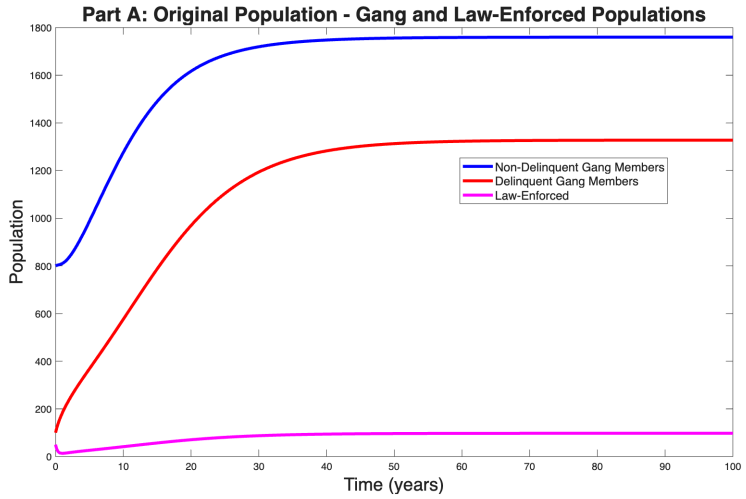


Figure 2: Subpopulations of gang-affiliated juveniles in the initial model using the original parameters from Lee and Do (2011

As shown in Figures 1 and 2, the susceptible population initially decreases while the non-delinquent and delinquent gang populations increase. The non-delinquent gang population has rapid growth during the early years of the simulation, while the law-enforced population also increases as the delinquent population grows.

Over time, all four populations approach equilibrium. At the end of the simulation, approximately 68.15% of the population is susceptible, 17.60% consists of non-delinquent gang members, 13.28% consists of delinquent gang members, and 0.98% is law-enforced.

3.2 Modeling Increased Socioeconomic Status

We next considered a population with higher socioeconomic conditions to see how a more favorable social environment affects juvenile delinquency. We reduced the peer-pressure delinquency rate (g_i) from 1.00 to 0.42, reflecting a lower likelihood of non-delinquent gang members becoming delinquent through peer influence. Additionally, we reduced the average time non-delinquent gang members remain in G from 313 years to 2 years, increasing the gang departure rate (r_G) from 0.30 to 0.50. These changes represent reduced peer influence and a greater tendency for gang members to return to the susceptible population in higher socioeconomic populations. All other parameters and initial conditions remained unchanged from the initial model.

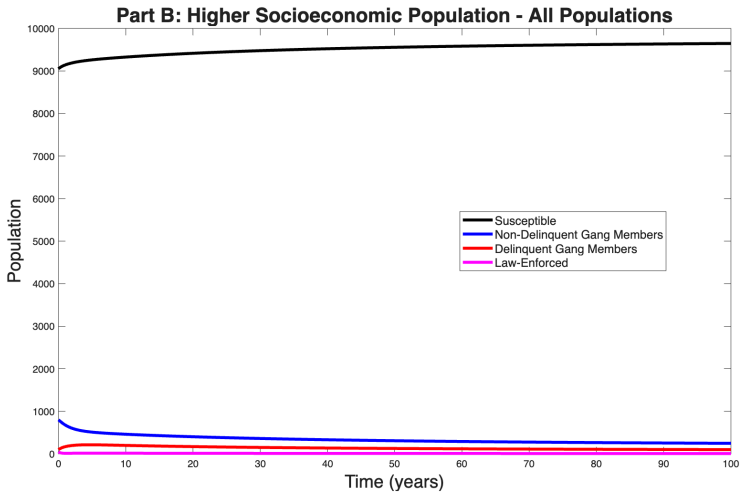


Figure 3: Population of susceptible and gang-affiliated juveniles in the higher socioeconomic model, incorporating reduced peer pressure and increased gang departure rates.

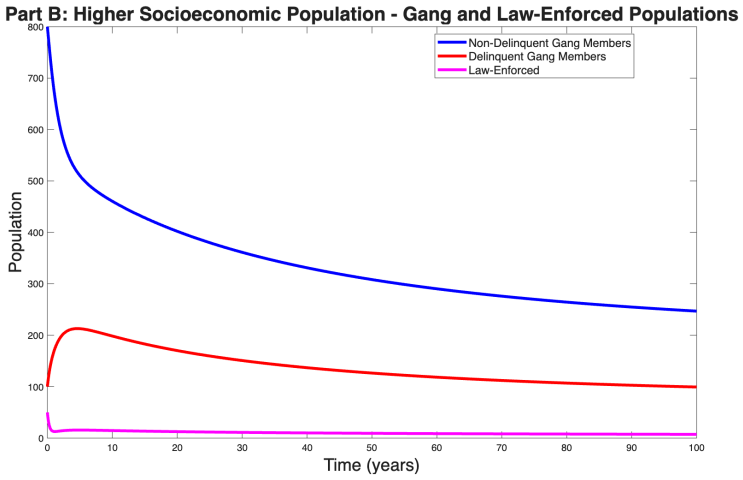


Figure 4: Subpopulations of gang-affiliated juveniles in the higher socioeconomic model

As shown in Figures 3 and 4, the higher socioeconomic model shows different trends compared to the initial model. The susceptible population remains relatively large and only increases, while the non-delinquent gang population decreases over time, never increasing. The delinquent population initially increases before gradually declining, and the law-enforced population follows a similar trend.

At the end of the 100-year simulation, approximately 96.46% of the population is susceptible, 2.47% consists of non-delinquent gang members, 0.99% consists of delinquent gang members, and 0.07% being law-enforced. Compared to the initial model, both non-delinquent and delinquent gang populations are significantly smaller, while the susceptible population is much larger. Although gang-affiliated populations continue to decrease toward the end of the simulation, their rates of decline have slowed.

3.3 Applying Memphis Parameters

We further apply our model to Memphis TN, a city with historically high violent crime rates and high rates of gang activity. This goes hand in hand with the high poverty rate in Memphis, especially among juveniles [8]. While we use the same parameters as in our initial model to describe the flows between populations by the assumption that these parameters describe general trends and therefore apply broadly, we make new estimates for the population proportions. Currently no precise estimates exist for juvenile gang membership and incarceration in Memphis, however estimates may be made using general gang violence statistics for Memphis and for the United States. The population of Memphis, according to the U.S. Census is around 609,647 for 2025 with an es-

timated 13% between the ages of 10 and 18 [8]. A very broad estimate for the total juvenile population may then be said to be around 79,254. If we take into account only those juveniles living below the poverty line, who may be more susceptible to gang influence, our total population is 30,750. An estimated 40% of urban gangs are juveniles between the ages of 5 and 17 [6]. The most recent estimate from the Tennessee Drug Threat Assessment puts the number of active gang members in Memphis at around 10000 [1]. Therefore, we estimate that the number of active juvenile gang members in Memphis, both delinquent and non-delinquent, to be about 4000. A total of 1,142 juveniles were held in detention in 2024 [3]. In 2026, the Memphis Safe Task Force reported that out of a total 13,140 arrests, 1,461 were of known gang members [5]. Assuming that this proportion is representative of overall trends in organized crime, we assume that a total of 127 juvenile gang members are incarcerated. In total, this results in a population that is 87% susceptible, 10% gang affiliated non-delinquent, 2.6% gang affiliated delinquent, and 0.4% arrested and detained.

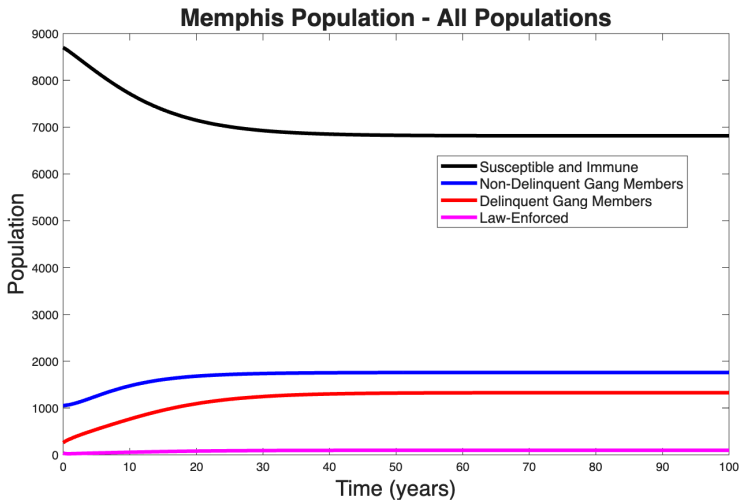


Figure 5: Populations of susceptible and gang-affiliated juveniles in Memphis.

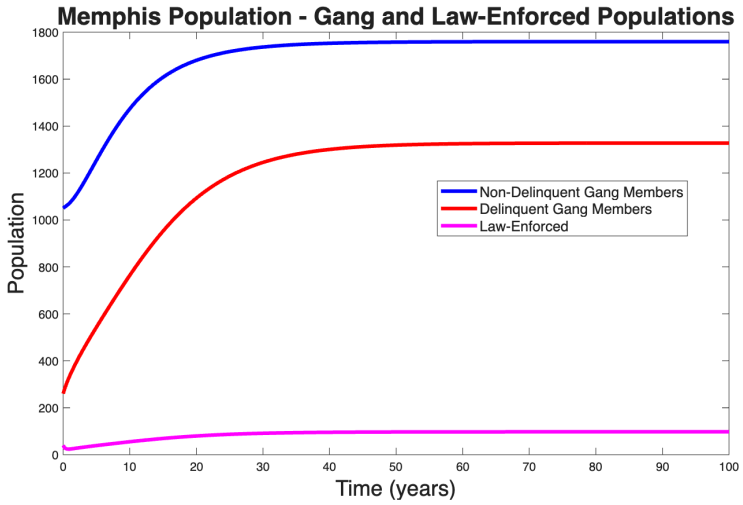


Figure 6: Subpopulations of gang-affiliated juveniles in Memphis.

Figure 5 and Figure 6 show the population trends using the population proportions estimated for Memphis. The final population proportions are as follows: 68.15% susceptible and immune, 17.60% non-delinquent gang-affiliated, 13.28% delinquent gang-affiliated, 0.98% law enforced. The populations estimated by this model follow the same general trend towards equilibrium as in model one, described in section 3.1. Likely as a result of the higher initial gang population, there is a higher final gang population than in the initial model which shows the importance of initial parameters to the long term model results. The trend towards equilibrium also shows that the flows into and out of each population are approximately equal so the overall values of each population become fixed as the model progresses in time.

3.4 Intervention Methods

In addition to considering the effects of different population parameters on our model, we consider the impacts of intervention methods. We use the population parameters and all flow parameters described in section 3.1, with the exception of the detainment rate i. We add an additional population described in 2.2 to model a population who has experienced early social intervention and is no longer susceptible to gang influence.

3.4.1 Increased Policing

The first intervention method we explore is increased policing, using a detainment rate of $i = 0.296$ and a small intervention rate $v = 0.02$ to model a situation in which policing is emphasized over social intervention.

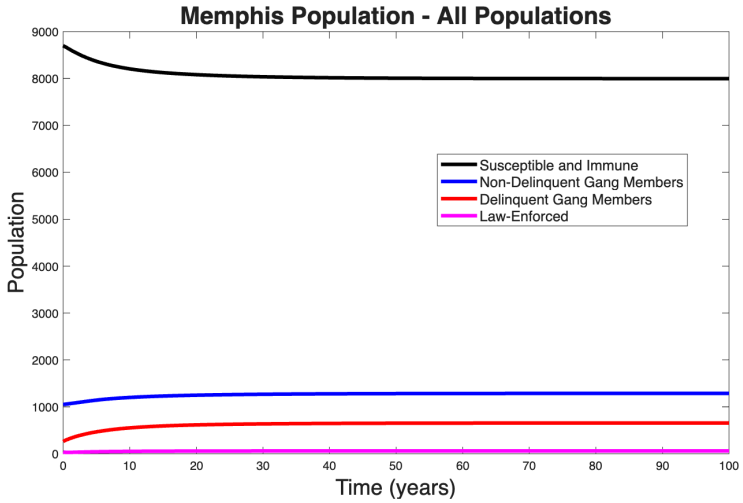


Figure 7: Populations of susceptible and gang-affiliated juveniles in Memphis under increased policing.

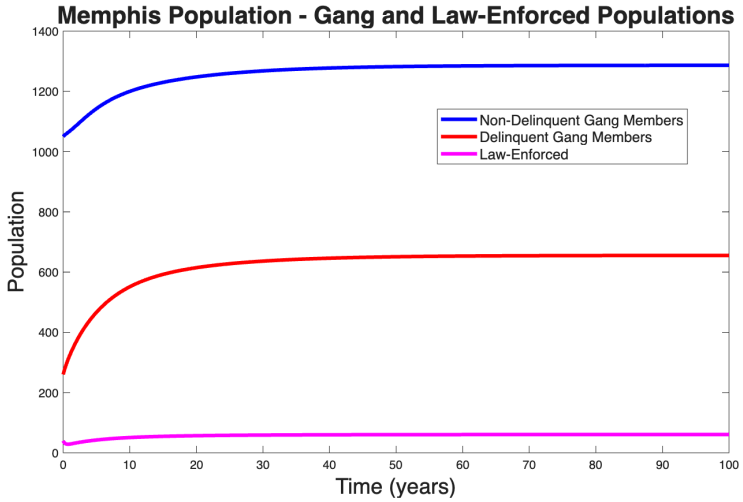


Figure 8: Subpopulations of gang-affiliated juveniles in Memphis under increased policing.

Figure 7 and Figure 8 show the trends in population proportions for models with increased policing. The final population proportions produced by these models are as follows: 79.98 % susceptible and immune, 12.87 % non-delinquent gang-affiliated, 6.55 % delinquent gang-affiliated, 0.61 % law enforced. While there is a notable decrease in the proportion of gang-affiliated juveniles in comparison to the original Memphis model, the populations still reach certain equi-

librium proportions after a certain amount of time, with a non-insignificant proportion of the population being gang-affiliated. This could suggest that while increased policing decreases overall gang activity, it does not directly impact the factors causing steady flows into and out of the gang-affiliated populations.

3.4.2 Early Intervention

The next intervention method we explore is aggressive early social intervention, using a detainment rate of $i = 0.235$ and a high intervention rate of $v = 0.10$ to model a situation in which members of the susceptible population are influenced away from gang activity and moved to an immune population who are no longer at risk of participating in gang activity.

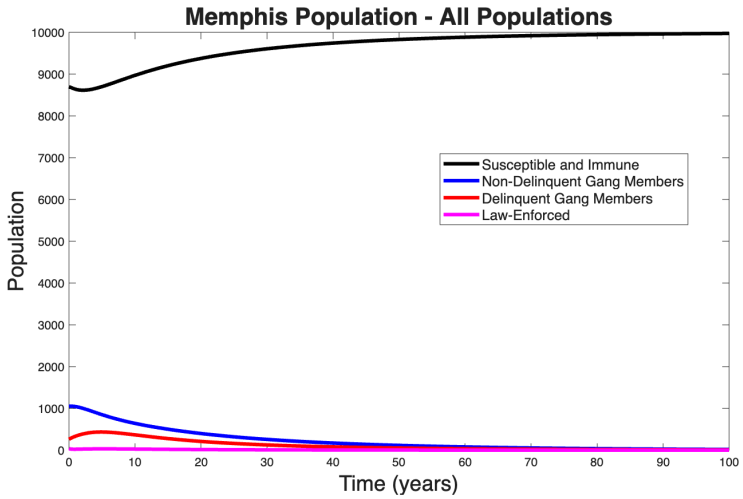


Figure 9: Populations of susceptible and gang-affiliated juveniles in Memphis under early intervention efforts.

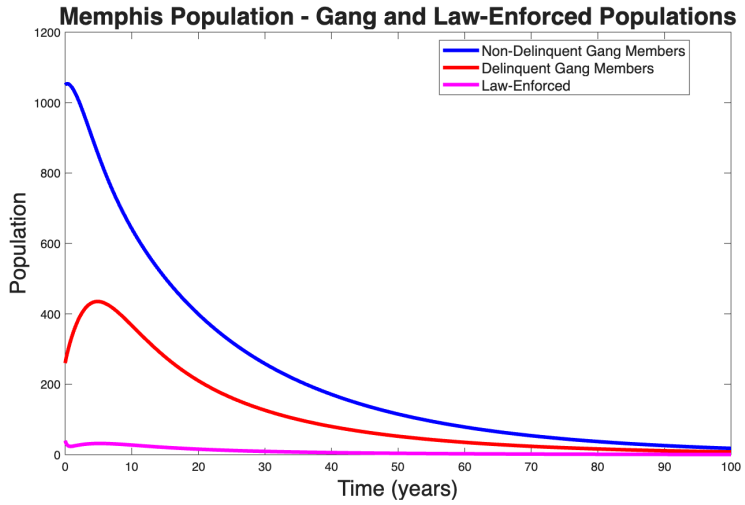


Figure 10: Subpopulations of gang-affiliated juveniles in Memphis under early intervention efforts.

Figure 9 and Figure 10 show the trends in population proportions for models with early social intervention. The final populations are as follows: 99.74 % susceptible and immune, 0.18 % non-delinquent gang-affiliated, 0.08 % delinquent gang-affiliated, and 0.006 % law enforced. In this model, we see that as members of the susceptible population are moved to an immune population, the population able to become gang-affiliated decreases steadily to zero, despite a high initial proportion of gang-affiliated juveniles. Rather than an approximately equal flow in and out of each population, the flow into gang populations appears to decrease with the decreasing susceptible population. This yields results similar to those observed in 3.2 where the populations of gang-affiliated juveniles decrease steadily as a result of the imbalance flows between the populations. This could imply that for total gang activity to decrease towards zero, sustained efforts must be made to prevent susceptible juveniles from being influenced to participate in gang activity.

4 Discussion

4.1 Ethical Considerations

It is important to seriously consider the potential ethical concerns inherent in these models of social phenomena. There are inherent limitations in modeling something as complex as human behavior with rigid and oversimplified mathematical models. Historically, oversimplified mathematical models have been used to harm marginalized communities, such as the communities most heavily impacted by gang violence. As such, it is important that when discussing the

results obtained from these models it is kept in mind that they are based on broad generalizations and should be interpreted as such.

4.2 Overall Trends

In general, we see that for groups of lower socioeconomic status, gang activity seems to reach an equilibrium after a certain amount of time, where flows into and out of gang-affiliated populations are approximately equal. This is true for both the initial and Memphis based model. This suggests that in these higher risk populations, gang activity becomes endemic, with the overall populations of gang-affiliated individuals neither increasing nor decreasing significantly. Even with an increase in policing, as modeled in 3.4.1, we see that while overall gang affiliation decreases with time, these populations still reach the equilibrium where the overall populations of gang-affiliated juveniles do not change.

In contrast, we see that in the model of higher socioeconomic groups, even with relatively high initial populations of gang members, the overall trend is a decrease in gang-affiliated populations. This suggests that with lower social incentive to join gangs and higher social incentive to leave them, the flow out of gangs is generally higher than the flow into them and these populations decline with time. We see this same trend when applying an early intervention program to these populations. Despite high initial gang populations, with sustained and aggressive intervention methods the population susceptible to gang influence becomes too small to sustain gang populations, and they decline with time. In both cases, it is through a sustained effort to reduce the flow into gangs that their overall populations decline and communities are protected from gang violence.

References

[1] National Drug Intelligence Center. Tennessee drug threat assessment. 2002.

[2] Young Lee and Tae Sug Do. A modeling perspective of juvenile crimes. *International Journal of Numerical Analysis and Modeling*, 2(4):369—-378, 2011.

[3] Memphis Police Department. 2024 annual report. Technical report, Memphis Police Department, 2024.

[4] U.S. Department of Justice. About violent gangs. 2002.

[5] Office of Public Affairs. Memphis safe task force commemorates first year of relentless action against violent crime. 2026.

[6] David Pyrooz. Gang membership between ages 5-17 in the united states. *Journal of Adolescent Health*, 56(4):414–419, 2015.

[7] Angela B. Shiflet and George W. Shiflet. *Introduction to Computational Science: Modeling and Simulation for the Sciences*. Princeton University Press, Princeton, NJ, 2nd edition, 2014.

[8] U.S. Census Bureau. Memphis city, tennessee, people. Technical report, U.S. Census Bureau, Washington, D.C, 2020.